MULTIDIMENSIONAL ODE (IVP)

Question for Day 7

1. Consider the following ODE-IVP

$$\frac{dy_1}{dt} = 9y_1 + 24y_2 + 5\cos(t) - \frac{1}{3}\sin(t)$$
$$\frac{dy_2}{dt} = -24y_1 - 51y_2 - 9\cos(t) + \frac{1}{3}\sin(t)$$

Initial condition: $Y_0 = [4/3 \quad 2/3]^T$

Make a computer program using a 4th order Runge Kutta technique to integrate this equation till large values of t. Use $h = 0.05$. Compare the results with the analytical solution

$$y_1 = 2e^{-3t} - e^{-39t} + \frac{1}{3}\cos(t)$$
$$y_2 = -e^{-3t} + 2e^{-39t} - \frac{1}{3}\cos(t)$$

2. Consider a study of isothermal plug flow reactor in which CO, NO and O2 (in automotive exhausts) react catalytically (iridium and rhodium). The dimensionless equations in absence of diffusion limitations are as follows:

$$\frac{dy_1}{dt} = kR(1-y_1)(1-y_2) + (S-R)(1-y_1)y_3$$
$$\frac{dy_2}{dt} = k(1-y_1)(1-y_2)$$
$$\frac{dy_3}{dt} = -(1-y_1)y_3$$

The initial conditions: $Y_0 = [0 \quad 0 \quad 1]^T$

y_1, y_2, y_3 are conversion of CO, NO, and O_2, respectively and t is a dimensionless reactor length. Integrate these for the following set of parameter (up to $t = 5$): $K = 2, R = 0.33, S = 0.33$

Use 4th order Runge Kutta method. Obtain results for different step size (h) till having step size does not make much difference in the value of y.

Analysis and Inputs

- Consider an ordinary differential equation (ODE) initial value problem (IVP):

$$\frac{d\mathbf{y}}{dt} = f(t, \mathbf{y}), \quad \mathbf{y}(t_0) = \mathbf{y}_0$$

  where $\mathbf{y} = [y_1, y_2, \dots, y_n]^T$ and $f(t, \mathbf{y}) = [f_1(t, \mathbf{y}), \dots, f_n(t, \mathbf{y})]^T$.

- According to Runge-Kutta 4th order method, the next value of $\mathbf{y}$ can be computed as:

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)$$

  where

$$\mathbf{k}_1 = \mathbf{f}(t_n, \mathbf{y}_n)$$

$$\mathbf{k}_2 = \mathbf{f}(t_n + \frac{h}{2}, \mathbf{y}_n + \frac{h}{2}\mathbf{k}_1)$$

$$\mathbf{k}_3 = \mathbf{f}(t_n + \frac{h}{2}, \mathbf{y}_n + \frac{h}{2}\mathbf{k}_2)$$

$$\mathbf{k}_4 = \mathbf{f}(t_n + h, \mathbf{y}_n + h\mathbf{k}_3)$$

- For the second question, to ensure that the chosen step size h does not affect the accuracy of the solution, the step size is repeatedly halved and the simulation is re-run at each halved value.

- The maximum absolute difference between the values of elements of $\mathbf{y}$ at the final reactor length obtained with successive step sizes is monitored, and halving is stopped once this difference falls below a user-specified tolerance, at which point the solution is taken to be independent of further step-size reduction.

- We read certain inputs from the user, defined in a file:

```
step_size_1: 0.05
step_size_2: 0.5
t_start: 0
t_end_1: 10
t_end_2: 5
```

```
F1 = { @(t,y)  9*y(1) + 24*y(2) + 5*cos(t) - sin(t)/3, ...
    @(t,y) -24*y(1) - 51*y(2) - 9*cos(t) + sin(t)/3 };

initial_guess_1 = [4/3, 2/3];

exact1 = { @(t)  2*exp(-3*t) -   exp(-39*t) + cos(t)/3, ...
    @(t) -  exp(-3*t) + 2*exp(-39*t) - cos(t)/3 };

K = 2; R = 0.33; S = 0.33;

F2 = { @(t, y) K * R * (1 - y(1)) * (1 - y(2)) + ...
        (S - R) * (1 - y(1)) * y(3), ...
    @(t, y) K * (1 - y(1)) * (1 - y(2)), ...
    @(t, y) - (1 - y(1)) * y(3) };

initial_guess_2 = [0, 0, 1];

[~, data] = readvars("input.txt", "TextType", "string");
step_size_1 = data(1); step_size_2 = data(2);
ti = data(3); tf_1 = data(4); tf2 = data(5);
```

Algorithm

- **Step 01. START**
- **Step 02.** Inputs:

Variable	Description
$\mathbf{f}$	$\frac{d\mathbf{y}}{dt} = f(t, \mathbf{y}) = [f_1(t, \mathbf{y}), \dots, f_n(t, \mathbf{y})]^T$
$\mathbf{y}_0$	Initial condition
t_0, t_f	Start and end time of the simulation
h	Fixed step size

- **Step 03.** Initialize: $t^{(0)} = t_0$, $\mathbf{y}^{(0)} = \mathbf{y}_0$, $n = 0$
- **Step 04.** Repeat until $t^{(n)} \geq t_f$:
  - **Step 09.** Update $\mathbf{y}^{(n+1)}$ from the aforementioned formula
  - **Step 10.** Update $t^{(n+1)} = t^{(n)} + h$
  - **Step 11.** Increment $n = n + 1$
- **Step 12.** End repeat {**Step 04**}
- **Step 13.** Outputs:

Variable	Description
$t^{(0..n)}$	Time history of guesses
$\mathbf{y}^{(0..n)}$	Solution guesses at those time steps

- **Step 14. STOP**

Implementation

```
%%file runge_kutta_4.m
function [t, Y_march] = runge_kutta_4(F, Y_0, t_0, t_t, h) %#ok<FNDEF>
    n = round((t_t - t_0) / h); t = t_0:h:t_t;
    Y_march = zeros(numel(F), n + 1);
    Y_march(:, 1) = Y_0;

    f = @(tt, yy) cellfun(@(g) g(tt, yy), F(:));

    for i = 1:n
        Y_current = Y_march(:, i); t_current = t(i);

        k1 = f(t_current, Y_current);
        k2 = f(t_current + h/2, Y_current + h*k1/2);
        k3 = f(t_current + h/2, Y_current + h*k2/2);
        k4 = f(t_current + h, Y_current + h*k3);

        Y_march(:, i + 1) = Y_current + h / 6 * (k1 + 2*k2 + 2*k3 + k4);
    end
end
```

```
File runge_kutta_4.m created successfully.
```

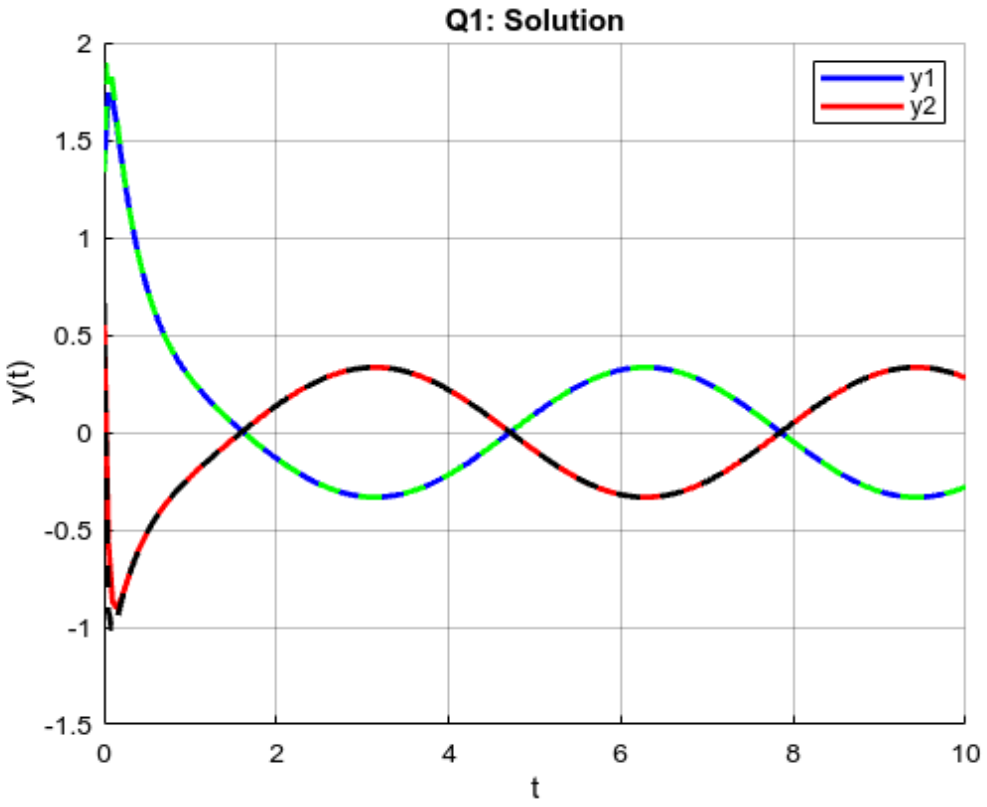
Problem 1

```
[t, Y] = runge_kutta_4(F1, initial_guess_1, ti, tf_1, step_size_1);

fprintf('Q1: h = %g, t = 0 to %g\n', step_size_1, tf_1);

figure('Color', 'w'); hold on; grid on;
set(gca, 'Color', 'w', 'XColor', 'k', 'YColor', 'k');
plot(t, Y(1,:), 'b-', 'LineWidth', 2);
plot(t, Y(2,:), 'r-', 'LineWidth', 2);
plot(t, exact1{1}(t), 'g--', 'LineWidth', 2);
plot(t, exact1{2}(t), 'k--', 'LineWidth', 2);
hold off; xlabel('t'); ylabel('y(t)');
title('Q1: Solution', 'Color', 'k');
legend('y1', 'y2', 'Location', 'best', 'TextColor', 'k', ...
    'Color', 'w', 'EdgeColor', 'k');
```

```
Q1: h = 0.05, t = 0 to 10
```



Problem 2

```
tol = 1e-8; h2 = step_size_2; prev_Y2 = inf;

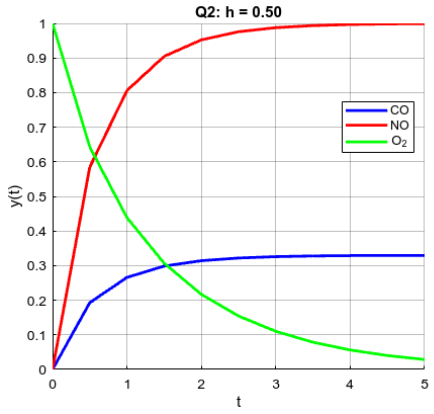
while true
    [~, Y2] = runge_kutta_4(F2, initial_guess_2, ti, tf2, h2);
    final_diff2 = max(abs(Y2(:, end) - prev_Y2(:, end)));
    if final_diff2 < tol; break; end
    prev_Y2 = Y2; h2 = h2 / 2;
end

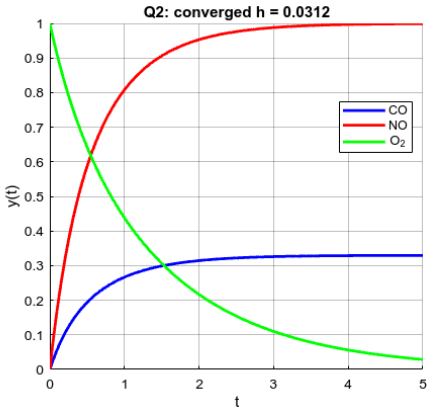
fprintf('Converged RK4 step size for Q2: h = %.4f\n', h2);
[t, Y2] = runge_kutta_4(F2, initial_guess_2, ti, tf2, h2);
[t_init, Y_init] = runge_kutta_4(F2, initial_guess_2, ti, ...
    tf2, step_size_2);

figure('Color', 'w', 'Position', [100, 100, 1100, 420]);
subplot(1,2,1); hold on; grid on;
set(gca, 'Color', 'w', 'XColor', 'k', 'YColor', 'k');
plot(t_init, Y_init(1,:), 'b-', 'LineWidth', 2); hold on;
plot(t_init, Y_init(2,:), 'r-', 'LineWidth', 2);
plot(t_init, Y_init(3,:), 'g-', 'LineWidth', 2);
hold off; xlabel('t'); ylabel('y(t)');
title(sprintf('Q2: h = %.2f', step_size_2), 'Color', 'k');
legend('CO', 'NO', 'O_2', 'Location', 'best', ...
    'TextColor', 'k', 'Color', 'w', 'EdgeColor', 'k');
grid on;

subplot(1,2,2); hold on; grid on;
set(gca, 'Color', 'w', 'XColor', 'k', 'YColor', 'k');
plot(t, Y2(1,:), 'b-', 'LineWidth', 2); hold on;
plot(t, Y2(2,:), 'r-', 'LineWidth', 2);
plot(t, Y2(3,:), 'g-', 'LineWidth', 2);
hold off; xlabel('t'); ylabel('y(t)');
title(sprintf('Q2: converged h = %.4f', h2), 'Color', 'k');
legend('CO', 'NO', 'O_2', 'Location', 'best', ...
    'TextColor', 'k', 'Color', 'w', 'EdgeColor', 'k');
grid on;
```

```
Converged RK4 step size for Q2: h = 0.0312
```





Discussion

In problem 1, a system of two coupled ordinary differential equations is solved numerically using the 4th-order Runge-Kutta (RK4) method. The initial conditions and parameters are defined, and the solution is computed over time using a fixed step size. At each time step, both the numerical and analytical solutions for $y_1(t)$ and $y_2(t)$ are calculated. Finally, the results are visualized through plots and the final values from both methods are printed for comparison to evaluate the accuracy of the numerical solution.

A similar thing was done in problem 2, but here we can obtain further information. At the reactor inlet, none of the CO or NO has reacted and O_2 is present at its full inlet value, so the initial condition is set as $y_1(0) = 0$, $y_2(0) = 0$ and $y_3(0) = 1$. As the dimensionless reactor length t increases from 0 to 5, the coupled reaction terms $(1 - y_1)(1 - y_2)$ and $(1 - y_1)y_3$ drive the system: y_1 and y_2 rise from zero as CO and NO are consumed, while y_3 falls from unity as O_2 is depleted.

To verify that the numerical solution is not sensitive to the chosen step size, the simulation was first performed with an initial step size h = 0.5, after which h was successively halved and the RK4 solution recomputed at each new step size. At every halving, the maximum absolute difference between the values of y_1, y_2 and y_3 at $t = 5$ obtained from the current and the previous step size was calculated.

The resulting conversion profiles of y_1 (CO), y_2 (NO) and y_3 (O_2) were plotted. All three curves display smooth, monotonic behaviour consistent with the underlying kinetics — y_1 and y_2 increase toward an asymptotic conversion as the t increases, while y_3 decreases correspondingly, reflecting the consumption of O_2 as the oxidation of CO and NO proceeds.